

Auditing DeepIV: Relevance and Reliability Signals for Nonparametric Instrumental-Variable Estimates

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Abstract

We study how to *audit* DeepIV estimates before trusting them, rather than certify identification. DeepIV flexibly handles nonlinear causal effects under endogeneity, but—unlike linear IV with its first-stage F -test—offers no operational warning when a fitted structural estimate should not be trusted [3]. We propose a stress-testing view that separates *distributional relevance* from *downstream reliability*. A Wasserstein relevance index ι measures how much the instrument shifts the treatment distribution; a family of reliability signals measures instability of the fitted structural function across random seeds and bootstrap resamples. In a controlled design where failure is known, relevance alone predicts failure only at chance level (AUC 0.43), while adding a reliability signal raises cross-validated AUC to 0.69 with a gain whose bootstrap confidence interval excludes zero. The recommendation is that a relevance statistic should never be reported alone as evidence a DeepIV estimate can be trusted.

Introduction

Endogeneity is pervasive in finance, where treatments such as prices, allocations, or exposures are jointly determined with outcomes. DeepIV [3] offers flexible counterfactual estimation, but unlike classical linear IV it provides no widely used weak-instrument diagnostic or operational warning that a fitted estimate is unreliable, especially when nonlinear first stages and ill-posed inverse problems interact [5]. A model can appear numerically stable in training yet remain unreliable for counterfactual use. We therefore frame the problem as *stress testing* rather than identification. Our contributions: a compact Wasserstein relevance index ι ; seed- and bootstrap-based reliability signals for fitted DeepIV structural functions; and a controlled evaluation of whether relevance-conditioned audits predict estimation failure better than relevance alone.

1 Background and Problem Formulation

Following DeepIV [3], let Y be an outcome, X an endogenous treatment, Z an instrument, and C controls, with $Y = h^*(X, C) + e$ and $\mathbb{E}[e \mid Z, C] = 0$, where

h^* is the causal response surface. DeepIV fits a first-stage treatment law $p_\phi(x \mid z, c)$ and then a structural function $\hat{h}$ by integrating a second-stage loss over the learned distribution. As in nonparametric IV, difficulty arises not only from instrument relevance but from the instability of an inverse problem [5].

Failure label. Given known ground truth, define

$$\text{Err}(\hat{h}, h^*) := \mathbb{E}_{\tilde{P}}[(\hat{h}(X, C) - h^*(X, C))^2], \quad \mathcal{F} := \{\text{Err} > \tau\}, \quad (1)$$

for a held-out $\tilde{P}$ and threshold $\tau > 0$. The audit predicts $\mathcal{F}$ before $\hat{h}$ is used.

Relevance index. For scalar or low-dimensional X ,

$$\iota := \mathbb{E}_C \mathbb{E}_{Z|C} \left[\frac{W_2^2(P_{X|C}, P_{X|Z, C})}{2 \text{Var}(X \mid C)} \right], \quad (2)$$

with W_2 the quadratic Wasserstein distance [11, 1]. It is outcome-agnostic—a relevance diagnostic, not a validity test.

Reliability signals. With $d(\hat{h}, \hat{h}') := (\mathbb{E}_{\tilde{P}}[(\hat{h} - \hat{h}')^2])^{1/2}$, independent seeds r on fixed data, and bootstrap resamples b ,

$$S_{\text{seed}} := \mathbb{E}_{r \neq r'}[d(\hat{h}_r, \hat{h}_{r'})], \quad S_{\text{boot}} := \mathbb{E}_{b \neq b'}[d(\hat{h}^{(b)}, \hat{h}^{(b')})], \quad (3)$$

$$S_{\text{gap}} := S_{\text{boot}} - S_{\text{seed}}, \quad S_{\text{rel}} := \frac{S_{\text{boot}} - S_{\text{seed}}}{S_{\text{seed}} + \varepsilon}, \quad S_{\text{res}} := S_{\text{boot}} - \hat{m}(\iota), \quad (4)$$

where $\hat{m}(\cdot)$ regresses S_{boot} on ι . These are candidate signals, not assumed to isolate identification error from optimisation noise.

2 Related Work

Weak-instrument diagnostics are well developed for linear IV, where the first-stage F -statistic and the Stock–Yogo critical values give an operational rule for when 2SLS should not be trusted [8, 9]. No comparable rule exists for DeepIV [3], whose first stage is a conditional density rather than a linear projection, so there is no finite coefficient vector to test. Our relevance index adapts the optimal-transport sensitivity theory of Borogonov et al. [1], which supplies boundedness, zero-independence and max-functionality for a Wasserstein-based index; we use it as a relevance measure rather

than a sensitivity measure. On the estimation side, van der Laan et al. [10] address many weak instruments in nonparametric IV, but propose an estimator rather than a diagnostic. Our reliability signals draw on the algorithmic-stability literature [2, 4, 7, 6], which assesses a fitted model by resampling; the contribution here is to separate resampling variation from optimisation variation and to condition both on relevance.

3 Experimental Setup

We evaluate the audit signals on a controlled data-generating process (DGP) in which the ground-truth structural function is known, so that the failure label $\mathcal{F}$ of (1) can be computed exactly. Simulation is essential rather than incidental: auditing a diagnostic requires knowing whether a fitted estimate is in fact unreliable, and observational data never reveals this.

Data-generating process. Let U be an unobserved confounder, Z an instrument, and e_x, e_y independent noise, all standard normal. With no controls ($C = \emptyset$),

$$X = \pi Z + U + e_x, \quad Y = h_\kappa(X) + \delta U + e_y, \quad (5)$$

with $h_\kappa(x) = \frac{1}{2}x + a \sin(\kappa x)$. Three parameters are varied independently.

Relevance π controls how strongly Z shifts X ; it is the quantity ι is intended to detect.

Ill-posedness κ controls the difficulty of the inverse problem. Since $X | Z=z \sim \mathcal{N}(\pi z, \sigma^2)$,

$$\mathbb{E}[\sin(\kappa X) | Z=z] = \sin(\kappa \pi z) e^{-\kappa^2 \sigma^2 / 2}, \quad (6)$$

so the high-frequency content of h_κ is exponentially damped in the observable conditional mean. With $\sigma^2 \approx 2$ the factor is 0.78 at $\kappa=0.5$ but 1.1×10^{-7} at $\kappa=4$; recovery then requires inverting a near-degenerate operator. This realises in closed form the singular-value decay that drives instability in nonparametric IV [5]. Because κ enters only through Y , the pair (X, Z) is unchanged by it, so any outcome-agnostic relevance measure is constant in κ by construction.

Confounding δ loads U onto Y only. This asymmetry is deliberate. If a single coefficient γ multiplies U in both equations, the resulting OLS bias is proportional to $\gamma/(\pi^2 + \gamma^2 + 1)$, which is bounded and non-monotone: raising γ inflates $\text{Var}(X)$ as fast as it adds covariance, so OLS never degrades and the design cannot motivate an IV method at all. Loading U on Y with a separate δ makes the bias proportional to $\delta/(\pi^2 + 2)$, which grows without bound. Table 1 confirms the corrected behaviour.

Estimators and metrics. The DeepIV pipeline uses a mixture-density first stage (5 Gaussian components) for $p(x | z)$ and a feed-forward second stage, trained by drawing S samples from the fitted treatment law and minimising the integrated second-stage loss. We benchmark against OLS (the confounded baseline) and 2SLS (linear IV). Errors are root-mean-square deviation

δ	OLS	2SLS	analytic bias $\delta/(\pi^2 + 2)$
0	0.282	0.280	0.000
1	0.217	0.280	0.167
2	0.203	0.280	0.333
4	0.334	0.280	0.667
6	0.539	0.280	1.000
8	0.761	0.281	1.333

Table 1: Normalised RMSE against confounding strength δ (concave shape, $\pi = 2$, $n = 4000$). OLS degrades without bound; 2SLS is unaffected.

from the true h_κ on a fixed grid $x \in [-3, 3]$, normalised by $\text{sd}(h_\kappa)$. Our implementation is deliberately compact and untuned; comparisons below should be read as statements about this pipeline, not about DeepIV at its best.

4 Results

4.1 The design makes endogeneity bite

Table 1 varies δ on the concave $\sqrt{\cdot}$ shape. OLS degrades monotonically once $\delta \geq 2$ and is more than twice as inaccurate as 2SLS by $\delta = 8$, while 2SLS is flat to three decimals: the IV estimators purge the U -channel, as intended. The design therefore satisfies the precondition for studying IV methods at all.

4.2 Sanity check, and a negative result on flexibility

Table 2 sweeps the structural shape at $\delta = 6$. The apparatus passes its sanity check: under a linear h , 2SLS attains 0.333 against DeepIV’s 1.458, so the correctly specified estimator wins decisively and the design does not spuriously favour the flexible model.

We also report a negative result that we believe is important to state plainly. Under confounding strong enough to break OLS, 2SLS outperforms our DeepIV pipeline on *every* shape tested, including the concave $\sqrt{\cdot}$ shape motivated by the empirical square-root law of price impact. Scanning $\delta \in \{1, 2, 3, 4\}$ we find no setting in which OLS is broken and DeepIV simultaneously improves on 2SLS: for $\delta \leq 3$ OLS is not yet degraded, and by $\delta = 4$ DeepIV has already fallen behind (0.355 vs 0.278). Increasing the sample to $n = 6000$ does not reverse this (0.353 vs 0.278). The interpretation is that a monotone, mildly curved h is well approximated by a line, so 2SLS pays little bias for a large variance advantage, while the flexible second stage must fit a target buried under δU noise.

This does not undermine the paper’s objective, because our contribution is an audit for practitioners who have already chosen DeepIV — typically for heterogeneity, controls, or dimensionality reasons absent from this univariate design — rather than a claim that DeepIV dominates linear IV. It does mean that a DeepIV-versus-2SLS superiority claim should not be made on the basis of these experiments.

h shape	OLS	2SLS	DeepIV
linear	1.976	0.333	1.458
mild sine	0.871	0.470	0.800
concave $\sqrt{\cdot}$	0.553	0.303	0.429
saturating tanh	0.810	0.441	0.726

Table 2: Normalised RMSE by structural shape ($\pi = 2$, $\delta = 6$, $n = 1200$, 8 seeds). The linear row is the sanity check. Under strong confounding 2SLS is the stronger estimator throughout for this compact DeepIV implementation.

		normalised Err at $\kappa =$				
π	ι	0.5	1.0	2.0	4.0	
0.1	0.018	1.71	1.73	3.66	3.10	
0.3	0.042	1.63	1.72	3.34	3.00	
0.6	0.106	1.70	1.82	4.44	4.43	
1.0	0.213	0.84	0.90	3.08	2.16	
2.0	0.439	0.86	1.02	3.81	2.56	

Table 3: Relevance index and estimation error over the $\pi \times \kappa$ grid ($\delta = 6$, $n = 1000$, mean of 3 seeds). Error rises sharply with ill-posedness while ι is flat along each row.

4.3 The relevance index behaves as intended

Across the grid, ι is a clean, monotone function of instrument strength (0.018, 0.042, 0.106, 0.213, 0.439 for $\pi = 0.1, \dots, 2.0$) and is invariant to κ up to sampling variation. Its constancy in κ is a structural fact rather than an empirical accident: ι depends only on (X, Z) , which κ leaves unchanged. This makes precise the sense in which a relevance diagnostic *cannot*, on its own, detect identification failure, and motivates the reliability signals of (4).

Table 3 shows why this matters operationally. Estimation error Err is driven mainly by κ and only weakly by π : the sample correlation of Err with κ is +0.45, against -0.16 with ι . Failure in this design is therefore largely an ill-posedness phenomenon, invisible to relevance.

4.4 Audit utility: do reliability signals add predictive power?

This is the test the framework exists to pass. We fit 60 independent DeepIV pipelines (5 relevance levels $\times$ 4 ill-posedness levels $\times$ 3 seeds). For each we record ι , the reliability signals of (3)–(4) ($R = B = 3$), and Err. The failure label is $\mathcal{F} = \{\text{Err} > \tau\}$ with τ the median error, giving 30 failures. We fit $\hat{m}(\iota)$ by regressing $\log S_{\text{boot}}$ on $\log \iota$ and form $S_{\text{res}} = S_{\text{boot}} - \hat{m}(\iota)$. Predictive performance is measured by leave-one-out cross-validated AUC, so no model is scored on its own training point, together with the false-pass rate at 80% sensitivity.

Three observations follow from Table 4. First, ι alone is *uninformative* about failure: its AUC of 0.434 is statistically indistinguishable from chance, which is the

audit model	LOO AUC	false-pass @ 80%
ι alone	0.434	0.733
$\iota + S_{\text{seed}}$	0.628	0.800
$\iota + S_{\text{boot}}$	0.653	0.533
$\iota + S_{\text{res}}$	0.662	0.533
$\iota + S_{\text{rel}}$	0.676	0.533
$\iota + S_{\text{gap}}$	0.688	0.567

Table 4: Audit utility. Relevance alone does not predict failure; every reliability signal improves on it. Best gain +0.253 AUC, bootstrap 95% CI [+0.072, +0.429], $\Pr(\text{gain} > 0) = 0.996$.

operational counterpart of the structural blindness established in §4.3. A relevance-only audit would provide false reassurance. Second, every reliability signal improves substantially on relevance alone, with the best combination reaching 0.688; the bootstrap 95% confidence interval for the gain excludes zero. This is direct evidence for the paper’s central proposition that relevance and reliability are distinct axes. Third, the strongest signals are those built on the *contrast* between resampling and seed variation, S_{gap} and S_{rel} , rather than either raw quantity; the relevance-residualised S_{res} performs comparably. Differences among the reliability signals are within sampling error at $n = 60$ and we do not claim an ordering among them.

Optimisation noise is first order. Seed variation is not a nuisance term. On the linear shape the DeepIV estimate has seed-induced standard deviation 0.075 against 0.003 for OLS, and across the grid S_{seed} is of the same order as S_{boot} (1.6–2.1 against 2.0–2.8). The fitted pipeline is therefore not a deterministic function of the data, and a reliability signal built from resampling alone conflates data sensitivity with optimisation noise. That the two are separately informative is precisely why S_{gap} and S_{rel} , which compare them, carry signal.

A named failure mode: high-frequency collapse.

At the most ill-posed settings the bootstrap replicates agree on a smooth estimate that omits the oscillatory component of h_{κ} entirely. Because the replicates agree on the *same* wrong function, S_{boot} can be small precisely where the estimate is worst. This is the degeneracy any stability-based audit must guard against: agreement among replicates certifies stability, not correctness, and a low reliability signal is informative only alongside adequate relevance.

5 Conclusion and Limitations

We have proposed auditing DeepIV rather than certifying it, using a Wasserstein relevance index together with reliability signals derived from seed and bootstrap variation. In a controlled design where failure is known, relevance alone predicts failure at chance level while adding a reliability signal raises AUC to 0.69 with a

gain whose confidence interval excludes zero. The practical recommendation is that a relevance statistic should never be reported alone as evidence that a DeepIV estimate can be trusted.

Several limitations qualify these results. The design is univariate and free of controls, so it exercises neither the heterogeneity nor the dimensionality that motivate DeepIV in practice. Our DeepIV implementation is compact and untuned, and in this design 2SLS outperforms it wherever confounding is strong enough to break OLS (§4.2); the audit results should therefore be read as statements about auditing a flexible pipeline, not as evidence of that pipeline’s superiority. The audit grid comprises 60 fitted pipelines with $R = B = 3$ replicates, so differences among the reliability signals are not resolved and the AUC estimates carry wide intervals. Bootstrap consistency for a deep pipeline is assumed rather than proven. Finally, and most importantly, none of these statistics tests instrument *validity*: the exclusion restriction remains an untestable assumption, and a high relevance score with a clean reliability profile says nothing about whether the instrument is legitimate. Extending the design to semi-synthetic financial data, where the treatment is order flow and the instrument is flow-induced trading, is the natural next step.

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